Mathematical Analysis of Hybrid ESKF-TCN Framework

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1 Boundedness of Observable Subspace

Lemma 1 (Semi-Definiteness of Information Matrix). *In the absence of global positional references (e.g., GPS), the absolute position and yaw states are unobservable. Consequently, the external information matrix $\Delta\mathcal{I}_{ext}$ provided by the TCN is **Positive Semi-Definite (PSD)**, not strictly positive definite over the full state space:*

$$\Delta\mathcal{I}_{ext} = H^T R_k^{-1} H \succeq 0 \tag{1}$$

*However, strictly positive information gain is guaranteed within the **observable subspace** Ω_{obs}, which comprises velocity, roll, pitch, and sensor biases. For any vector $v \in \Omega_{obs}$, the quadratic form satisfies $v^T \Delta\mathcal{I}_{ext} v \geq \delta\|v\|^2 > 0$.*

1.1 Architecturally Constrained Uncertainty Modeling

The heteroscedastic uncertainty is predicted via a Temporal Convolutional Network (TCN). The diagonal measurement noise covariance matrix R_k is defined as:

$$R_k = \mathrm{diag}(\sigma_{v,k}^2, \sigma_{\omega,k}^2) \tag{2}$$

Crucially, to satisfy the boundedness assumption required for stability analysis, each variance $\sigma_{i,k}^2$ is constrained within a physical range $[\sigma_{min}^2, \sigma_{max}^2]$ via a **hard-clamping activation layer** at the network output. This architectural design ensures numerical stability regardless of the network's weights:

$$0 < \lambda_{min}\mathbf{I} \leq R_{min} \leq R_k \leq R_{max} < \infty \tag{3}$$

1.2 Continuous Pseudo-Measurement Injection

Unlike traditional ZUPT approaches that only correct errors during static phases, the proposed framework employs the TCN as a **continuous virtual sensor**. The network continuously predicts the pseudo-velocity vector and gravity vector, ensuring that the measurement update loop remains closed even during dynamic motion. This continuous injection provides the necessary information flow to bound the drift of the observable states.

2 Divergence of Pure Inertial Integration

2.1 Error State Dynamics

Let $\delta\mathbf{x} \in \mathbb{R}^n$ denote the error state vector, which includes errors in position, velocity, orientation, and sensor biases. In the absence of external corrections (pure integration), the discrete-time evolution of the error state is governed by:

$$\delta\mathbf{x}_{k+1} = \mathbf{F}_k \delta\mathbf{x}_k \tag{4}$$

where $\mathbf{F}_k$ represents the discrete-time state transition matrix derived from the system dynamics.

2.2 Information Decay and Rank Deficiency

The Fisher Information Matrix (FIM), denoted as $\mathcal{I}_k$, represents the total amount of information the system holds regarding its states. The recursive update of the FIM is given by:

$$\mathcal{I}_{k+1} = \mathbf{F}_k^{-T}\mathcal{I}_k\mathbf{F}_k^{-1} + \mathbf{H}_{k+1}^T\mathbf{R}_{k+1}^{-1}\mathbf{H}_{k+1} \tag{5}$$

In a pure inertial navigation scenario, there are no external measurements available, implying $\mathbf{H}_k = \mathbf{0}$. Consequently, the information update term vanishes:

$$\mathbf{H}_k^T\mathbf{R}_k^{-1}\mathbf{H}_k = \mathbf{0} \tag{6}$$

As a result, the FIM evolves purely based on system dynamics. Due to the inherent geometry of inertial sensors, states corresponding to the global position (δp) and yaw ($\delta\psi$) form the null space of the observation manifold. This leads to a rank deficiency in the information matrix:

$$\text{rank}(\mathcal{I}_k) < \dim(\delta\mathbf{x}) \tag{7}$$

2.3 Unbounded Drift Theorem

Since the error covariance matrix $\mathbf{P}_k$ is the inverse of the FIM ($\mathbf{P}_k = \mathcal{I}_k^{-1}$), the rank deficiency of $\mathcal{I}_k$ implies that the eigenvalues corresponding to the unobservable subspace diverge to infinity over time:

$$\lim_{t\to\infty} \lambda_{max}(\mathbf{P}_k) = \infty \tag{8}$$

This formally proves that open-loop inertial integration inevitably leads to unbounded drift in unobservable states.

3 Reduction of CRLB via Hybrid Information Injection

3.1 Augmented Information Matrix

Let $\mathcal{I}_{base,k}$ denote the Fisher Information Matrix (FIM) of the baseline pure inertial navigation system. In the proposed hybrid architecture, the information matrix is augmented by external pseudo-measurements provided by the TCN-ESKF coupling:

$$\mathcal{I}_{hybrid,k} = \mathcal{I}_{base,k} + \Delta\mathcal{I}_{ext,k} \tag{9}$$

The external information gain, $\Delta\mathcal{I}_{ext,k}$, is derived from the pseudo-velocity observation equation:

$$\Delta\mathcal{I}_{ext,k} = \mathbf{H}^T\mathbf{R}_k^{-1}\mathbf{H} \tag{10}$$

3.2 Probabilistic Measurement Covariance

The measurement update logic is governed by a switching mechanism based on the zero-velocity probability $p_k \in [0, 1]$. The adaptive covariance $\mathbf{R}_k$ is defined as a convex combination of high and low uncertainty states:

$$\mathbf{R}_k = (1 - p_k)\mathbf{R}_{high} + p_k\mathbf{R}_{low} \tag{11}$$

Since $\mathbf{R}_{high}$ and $\mathbf{R}_{low}$ are symmetric positive definite (SPD) matrices, their convex combination $\mathbf{R}_k$ is strictly positive definite for all p_k. This ensures that the information gain $\Delta\mathcal{I}_{ext,k}$ is non-zero and bounded.

3.3 Cramér-Rao Lower Bound (CRLB) Reduction

The Cramér-Rao Lower Bound, which defines the theoretical minimum variance of any unbiased estimator, is the inverse of the FIM. By the Loewner order property of positive definite matrices ($A \succeq B \Rightarrow A^{-1} \preceq B^{-1}$), the addition of external information yields:

$$(\mathcal{I}_{base,k} + \Delta\mathcal{I}_{ext,k})^{-1} \preceq \mathcal{I}_{base,k}^{-1} \tag{12}$$

Consequently, the hybrid architecture strictly reduces the error lower bound:

$$CRLB_{hybrid} \leq CRLB_{base} \tag{13}$$

This probabilistic decay mechanism acts as a differentiable soft constraint, effectively minimizing the CRLB in the zero-velocity regime without introducing algorithmic discontinuities.

4 Partial Lyapunov Stability Analysis

4.1 Definition of Observable Error State

Standard Lyapunov analysis on the full state vector $\delta\mathbf{x}$ yields a contradiction because the absolute position error is physically unbounded without global references (e.g., GPS). To resolve this, we partition the error state and define the **Observable Error State** vector $\delta\mathbf{x}_{obs}$:

$$\delta\mathbf{x}_{obs} = [\delta\mathbf{v}^T, \delta\phi, \delta\theta, \delta\mathbf{b}_a^T, \delta\mathbf{b}_g^T]^T \tag{14}$$

This subspace Ω_{obs} excludes the unobservable global position $\delta\mathbf{p}$ and yaw $\delta\psi$, focusing only on states that are theoretically observable via the TCN's pseudo-measurements (velocity and gravity vectors).

4.2 Lyapunov Function Candidate

We define a Lyapunov function candidate V_k exclusively on this observable manifold. To avoid circular reasoning associated with covariance-weighted norms, we employ the squared Euclidean norm:

$$V_k(\delta\mathbf{x}_{obs,k}) = \|\delta\mathbf{x}_{obs,k}\|^2 = \delta\mathbf{x}_{obs,k}^T\delta\mathbf{x}_{obs,k} \tag{15}$$

This scalar represents the "correctable energy" of the system—the magnitude of errors that the hybrid architecture is capable of suppressing.

4.3 Energy Expansion in Prediction Step

During the prediction step, the error dynamics are governed by the system transition matrix $\mathbf{F}$. Within the observable subspace, the energy expands due to process noise and linearization errors:

$$V_{k|k-1} \leq \|\mathbf{F}_{obs,k-1}\|^2 V_{k-1|k-1} + \epsilon = \alpha^2 V_{k-1|k-1} + \epsilon \tag{16}$$

where $\alpha \approx 1$ represents the expansion factor and ϵ denotes the bounded process noise energy.

4.4 Energy Dissipation in Update Step

The TCN continuously injects pseudo-velocity and gravity observations. Since these measurements directly map to the observable subspace Ω_{obs}, the observation matrix $\mathbf{H}_{obs}$ is full-rank. The update step induces a strict energy contraction:

$$V_{k|k} \leq \|\mathbf{I} - \mathbf{K}_k\mathbf{H}_{obs}\|^2 V_{k|k-1} = \rho^2 V_{k|k-1} \tag{17}$$

Crucially, because the information matrix is positive definite in this subspace (Lemma 1), the contraction factor satisfies $\rho < 1$.

5 Stability Synthesis and Drift Dynamics

5.1 Global Convergence Condition

Combining the expansion and contraction phases, the evolution of the correctable energy is given by:

$$V_{k|k} \leq \rho^2 \alpha^2 V_{k-1|k-1} + \rho^2 \epsilon \tag{18}$$

For the observable states to remain stable, the condition $\gamma = \rho\alpha < 1$ must be satisfied. This inequality mathematically dictates that the information injection rate provided by the TCN must overpower the natural drift rate of the IMU.

5.2 UUB of Observable States

Under the condition $\gamma < 1$, the energy V_k converges to a bounded region defined by the noise floor. Consequently, the velocity, tilt, and bias errors are **Uniformly Ultimately Bounded (UUB)**:

$$\lim_{k \to \infty} E[\|\delta \mathbf{x}_{obs,k}\|^2] \leq \frac{\rho^2 \epsilon}{1 - \gamma^2} \tag{19}$$

This proves that the dynamic properties of the trajectory do not diverge.

5.3 Linear Dispersion Dynamics via Random Walk Theory

While the observable states (velocity, attitude) are proven to be UUB, the global position error $\delta \mathbf{p}_k$ evolves as the temporal integration of the velocity error. Here, we analyze the growth rate of this integration to demonstrate the suppression of cubic drift.

5.3.1 Discrete-Time Error Accumulation

The position error update equation in the discrete-time domain is given by:

$$\delta \mathbf{p}_k = \delta \mathbf{p}_{k-1} + \delta \mathbf{v}_{k-1} \Delta t \tag{20}$$

Expanding this recursively from the initial state ($k = 0$) yields the summation form:

$$\delta \mathbf{p}_k = \delta \mathbf{p}_0 + \Delta t \sum_{i=0}^{k-1} \delta \mathbf{v}_i \tag{21}$$

5.3.2 Stochastic Growth Rate (Random Walk)

From the previous stability analysis (Section 5.2), the velocity error $\delta \mathbf{v}_i$ is a stationary process with bounded covariance Σ_v (due to UUB). Assuming the residual velocity errors are uncorrelated over long intervals (white noise approximation due to TCN corrections), the variance of the position error grows as:

$$E[\|\delta \mathbf{p}_k\|^2] \approx (\Delta t)^2 \sum_{i=0}^{k-1} E[\|\delta \mathbf{v}_i\|^2] \approx (\Delta t)^2 \cdot k \cdot \mathrm{tr}(\Sigma_v) \tag{22}$$

Substituting total time $t = k\Delta t$, the root-mean-square (RMS) error follows a Random Walk profile:

$$\mathrm{RMS}(\delta \mathbf{p}(t)) \propto \sqrt{t \cdot \Delta t \cdot \mathrm{tr}(\Sigma_v)} \propto O(\sqrt{t}) \tag{23}$$

5.3.3 Deterministic Worst-Case Bound (Linear Drift)

In the worst-case scenario where the TCN exhibits a non-zero residual mean error (constant bias $\boldsymbol{\mu}_v$) within the bounded region $\|\boldsymbol{\mu}_v\| \leq \nu_{max}$, the drift becomes linear:

$$\|\delta \mathbf{p}(t)\| \leq \int_0^t \|\delta \mathbf{v}(\tau)\| d\tau \leq \int_0^t \nu_{max} d\tau = \nu_{max} \cdot t \tag{24}$$

Consequently, the system guarantees a drift rate of **$O(t)$ (Linear)**. This stands in stark contrast to pure inertial integration, where uncorrected acceleration bias $\mathbf{b}_a$ leads to double integration errors growing as $O(t^2)$ or $O(t^3)$.

6 Conclusion

6.1 Restatement of Premises

The theoretical validity of the Trajecto framework rests upon the following core premises:

1. **Bounded Measurement Noise**: Sensor uncertainties are strictly confined within stable numerical ranges via architectural clamping.
2. **Continuous Pseudo-Observability**: The TCN provides continuous predictions of velocity and gravity vectors, ensuring that the Kalman update loop remains closed even during dynamic motion.
3. **Observable Subspace Stabilization**: While absolute position remains unobservable without GPS, the hybrid coupling effectively stabilizes the observable subspace (velocity and tilt), preventing the catastrophic exponential divergence typical of low-cost IMUs.

6.2 Final Verdict

By combining the reduction of the Cramér-Rao Lower Bound (CRLB) with the proof of UUB stability for dynamic states, this work establishes that the proposed hybrid ESKF-TCN architecture theoretically outperforms pure inertial baselines. The system achieves a robust convergence where deep learning-driven innovations constrain the chaotic IMU drift. Crucially, it transforms the mathematically inevitable position drift from an exponential explosion into a manageable linear accumulation, enabling precise centimeter-level handwriting reconstruction.